

Sports Analysis (Olympic Games)

Problem Statement:

The objective of this analysis is to gain valuable insights into the historical data of the Olympic Games by exploring various dimensions, including games, sports, events, participants, medals, and regional representation. The scope encompasses data from multiple editions of the Olympics, aiming to identify trends, patterns, and significant insights from different aspects of the Games. The primary goal is to provide a detailed overview of the Olympics' history and evolution, and the analysis will generate recommendations to enhance future Olympics' organization, sports selection, and representation of diverse regions. The final deliverable will be a comprehensive report and presentation showcasing the findings and historical significance of the Olympic Games.

Dataset Description:

The dataset here is a comprehensive collection of information about the Olympic Games, offering a detailed view of various aspects related to this prestigious international sporting event. It encompasses data from multiple interconnected tables, providing valuable insights into the history, sports, participants, and locations of the Olympic Games.

Table Explanations:

Sport Table:

This table contains a comprehensive list of all sports featured in the Olympics, covering both the summer and winter editions. Each sport is further categorized into specific events, catering to a wide range of athletic disciplines.

Event Table

The Event table offers detailed information about the various events held within each sport. Events are categorized based on gender, with separate entries for Men's, Women's, and Mixed events.

City Table

This table provides an extensive list of cities from around the world that have hosted or participated in the Olympic Games over the years.

Games Table

The Games table records every instance of the Olympic Games since its inception in 1896. It includes essential details such as the year in which the Games were held, as well as distinguishing between Summer and Winter Olympics.

Games City Table

This joining table establishes the relationship between the Games and City tables, capturing instances where the Olympic Games were jointly hosted by multiple cities, as seen in the case of the 1956 Olympics held in both Melbourne and Stockholm.

NOC Region Table

This table consists of NOC codes, representing National Olympic Committees, and their corresponding countries. It acts as a reference to identify the countries participating in the Olympic Games.

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Person Table

The Person table records information about individuals who have competed in the Olympics. It includes their names, gender, height, and weight, with height and weight data remaining consistent across different Olympic editions.

Person Region Table

This joining table establishes connections between individuals and the countries they represented while competing in the Olympics. It accommodates scenarios where a person may have competed for multiple countries in different Olympic Games.

Games Competitor Table

This table serves as a joining table, linking individuals to specific Olympic Games, showcasing the participants and their involvement in each event.

Medal Table

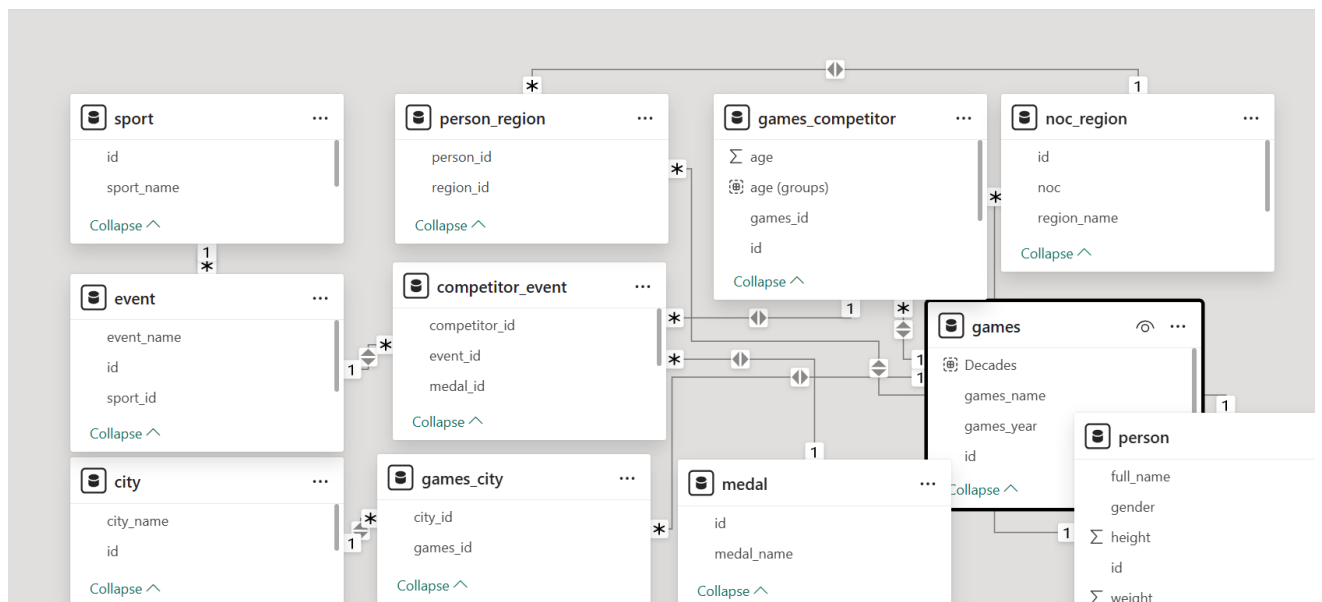
A small table listing the different types of medals awarded at the Olympics, including Gold, Silver, Bronze, and N/A (indicating no medal).

Competitor Event Table

This table represents the largest one, providing crucial details about the combination of competitors, the events they participated in, and the medals they received, if any.

ER Diagram:

Based on the above datasets there has been some relation between tables and those relations are established through Entity-Relationship-Diagram below.



The cardinality and table relations are shown as 1 to Many or Many to 1 with cross filter direction are shown in the below diagram for better understanding of table structures and its relations.

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From: table (column) ↑	Relationship	To: table (column)
competitor_event (competitor...		games_competitor (id)
competitor_event (event_id)		event (id)
competitor_event (medal_id)		medal (id)
event (sport_id)		sport (id)
games_city (city_id)		city (id)
games_city (games_id)		games (id)
games_competitor (games_id)		games (id)
games_competitor (person_id)		person (id)
person_region (person_id)		person (id)
person_region (region_id)		noc_region (id)

MECE breakdown:

No	Analysis Area	Short Explanation
1	Games Evolution	Analyses the growth and development of the Olympic Games over time, including editions, seasons, host cities, and expansion of sports/events.
2	Sports & Events	Examines the structure and diversity of Olympic sports and events, including newly introduced, discontinued, and gender-based events.
3	Athlete Demographics	Studies participant characteristics such as gender, age groups, height, weight, and athlete participation

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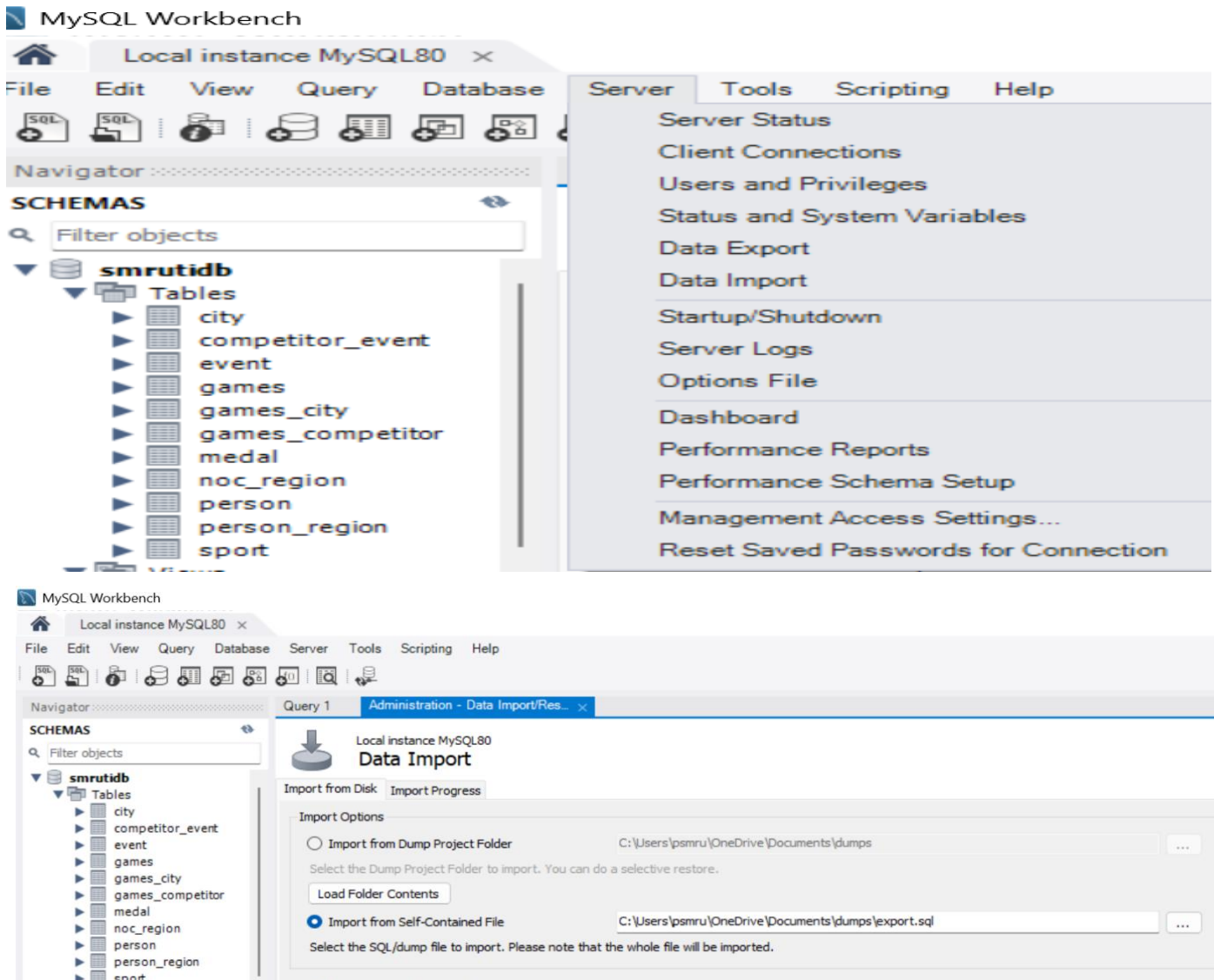
		trends across different Olympics.
4	Country Performance	Evaluates country-wise participation and success by analysing medals, athlete representation, and dominance across sports and Olympic editions.
5	Medal Analysis	Focuses on medal distribution patterns across countries, sports, genders, and Olympic years to identify competitive performance trends.
6	Geographic/Host Analysis	Investigates hosting patterns, regional representation, and the geographical distribution of Olympic Games and participating nations.
7	Participation vs Performance	Compares athlete participation levels with medal achievements to measure efficiency and competitive success of countries and sports.
8	Gender & Diversity	Analyses gender inclusion and diversity trends by studying male, female, and mixed participation and event representation.
9	Historical Trends	Identifies long-term trends and changes in participation, medals, sports growth, and regional performance across Olympic history.
10	Predictive Insights	Provides advanced analytical insights and future-oriented observations using historical Olympic data and performance patterns.

Data Connection and access:

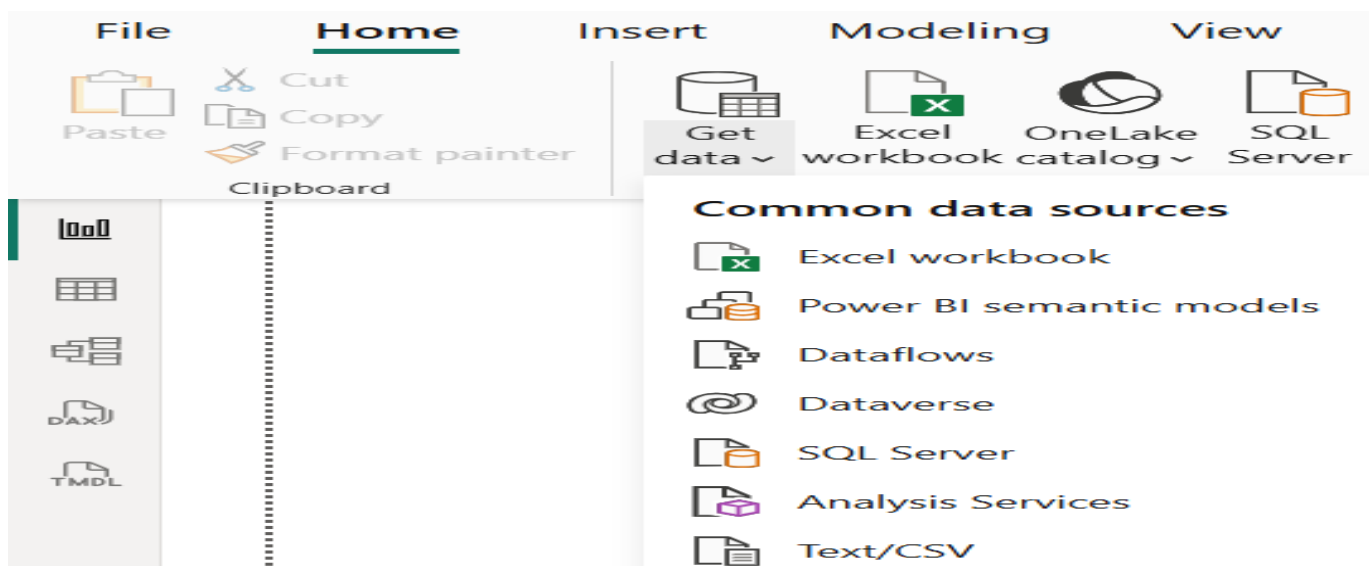
To achieve the above problem statements, I have imported the SQL scripts of the whole dataset to MySQL workbench through below path:

1. **MySQL Workbench**
2. **Server**
3. **Import Data**
4. **Import From Self Contained File**

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Similarly CSV files are imported to PBI Dashboard through GET Data and then text/csv.



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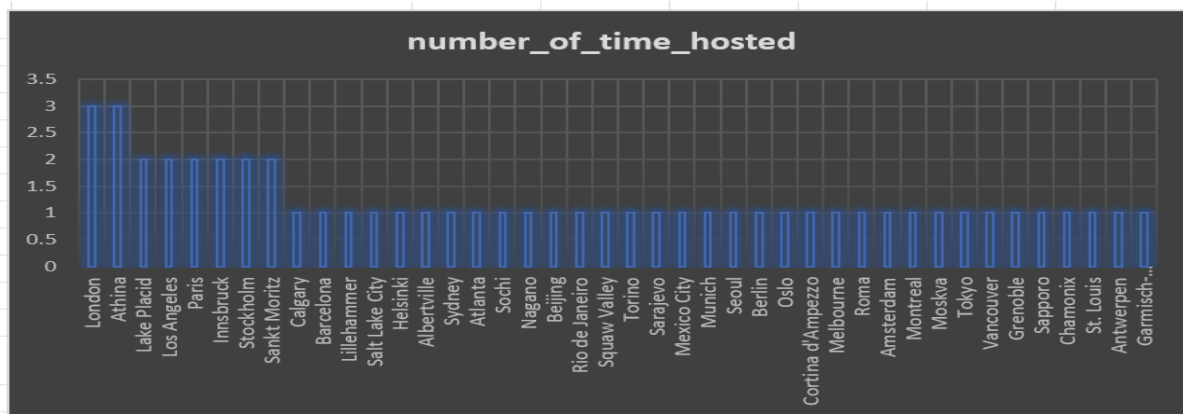
Exploratory DA Analysis (By MySQL Workbench and EXCEL):

1.Are there any trends or patterns in the frequency of hosting Olympic Games?

To establish this problem statement, I have written a SQL query by joining games, game city and city tables with their respective common relation and grouping and ordering the result with cityname and number of time hosted respectively.

```
select c.city_name,count(*) number_of_time_hosted
      from games g
      join games_city gc
      on g.id=gc.games_id
      join city c
      on gc.city_id=c.id
      group by c.city_name
      order by number_of_time_hosted desc;
```

So, from this I found a particular pattern in which it clearly shows that London and Athina city hosted maximum no of times (3). Lake Placid, Los Angeles, Paris, Innsbruck, Stockholm and Sankt Moritz hosted 2 times each with rest cities got single chance to host Olympics game.



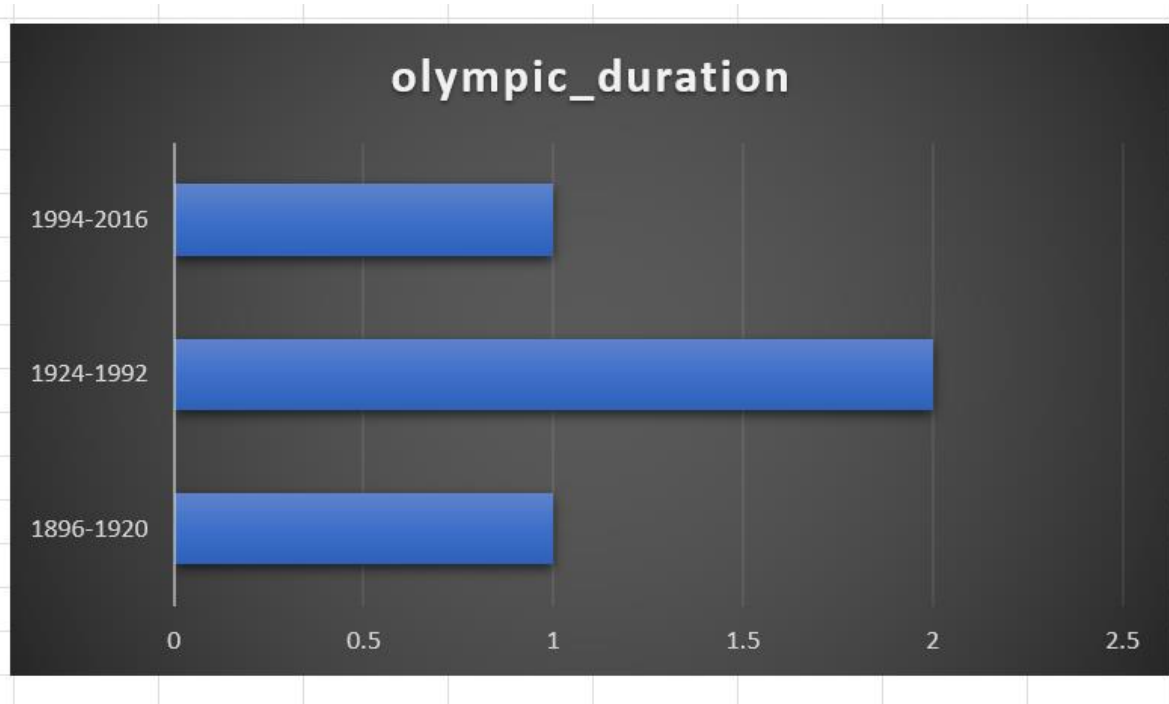
2. How has the duration of Olympic Games changed over time?

To achieve this problem statement, I have just grouped and sorted the games table by games year.

```
select games_year, count(*) olympic_duration
      from games
      group by games_year
      order by games_year;
```

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So from the above query we got that from 1896 to 1920 and 1994-2016 there has been single edition of Olympics games but from 1924-1992 two Olympics events are held.



3. Are there any notable events or occurrences associated with specific Olympic Games?

To achieve this problem statement, I have joined the competitor event, event and medal tables with their respective common relations followed by grouping the result through event name and sorting by no of participants.

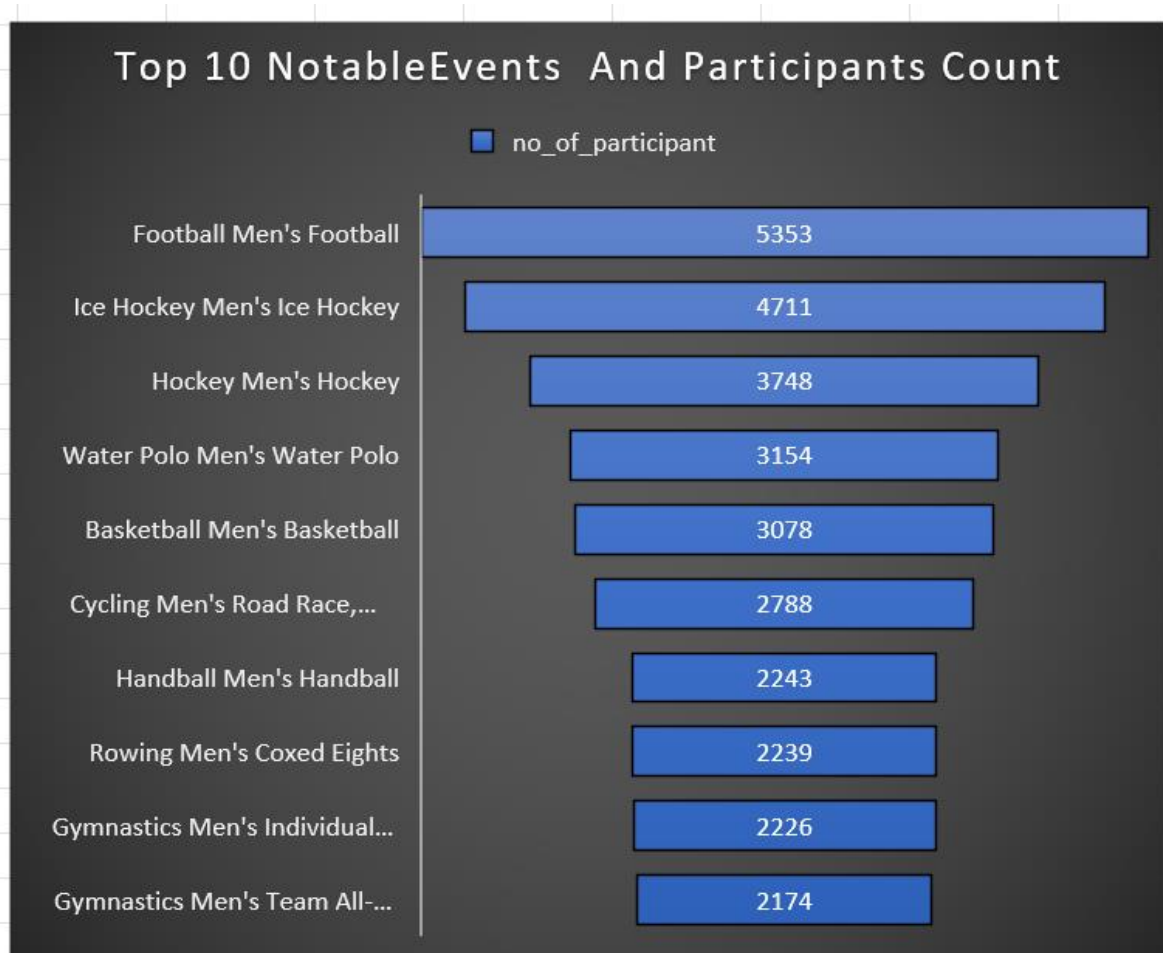
```
select e.event_name,count(ce.competitor_id) no_of_participant
from competitor_event ce
join event e
on ce.event_id=e.id
join medal m
on ce.medal_id=m.id
group by e.event_name
order by no_of_participant desc;
```

So, we got that Men's football, Men's Ice Hockey, and Men's Hockey are notable events because in these events there has been highest number of participants are there with count 5353,4711 and 3748 respectively.

In Gymnastic Men's Individual and team events there has been less participants registered as count 2226 and 2174 respectively.

So these events are not considered as notable events. The result table is listed below.

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4. Are there any emerging sports that have been recently added to the Olympics

To achieve this problem statement, I had to join competitor event, games competitor, games, event and sports table with their related columns, then grouping the result by sports name along with sorting by year of introduce.

```
select s.sport_name,min(g.games_year) year_of_introduce,
count(ce.competitor_id) participants_per_sport
from competitor_event ce
join games_competitor gc
on ce.competitor_id=gc.id
join games g
on gc.games_id=g.id
join event e
on e.id=ce.event_id
join sport s
on e.sport_id=s.id
group by s.sport_name
order by year_of_introduce desc;
```


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From the above query we got that a new emerging sport has been introduced in recent Olympic games named as Rugby Sevens. About 300 participants participated in these sports with multiple sporting events so we can consider it as emerging.

sport_name	year_of_introduce	participants_per_sport
Rugby Sevens	2016	299
Triathlon	2000	529
Trampolining	2000	152
Taekwondo	2000	606

5. How has the popularity of certain sports changed over the years?

To achieve this problem statement, I had to join competitor event, games competitor, games, events, and sports with their related column, followed by where clause between 1896 to 1956 and 1957-2016 and grouping and sorting by sport name.

For 1st decade:

```
SELECT
    s.sport_name,
    COUNT(ce.competitor_id) AS total_competitors
FROM competitor_event ce
JOIN games_competitor gc
ON ce.competitor_id = gc.id
JOIN games g
ON gc.games_id = g.id
JOIN event e
ON e.id = ce.event_id
JOIN sport s
ON e.sport_id = s.id
WHERE g.games_year BETWEEN 1896 AND 1956
GROUP BY s.sport_name
ORDER BY sport_name;
```

For 2nd decade:

```
SELECT
    s.sport_name,
    COUNT(ce.competitor_id) AS total_competitors
FROM competitor_event ce
JOIN games_competitor gc
ON ce.competitor_id = gc.id
JOIN games g
ON gc.games_id = g.id
JOIN event e
ON e.id = ce.event_id
JOIN sport s
ON e.sport_id = s.id
WHERE g.games_year BETWEEN 1957 AND 2016
GROUP BY s.sport_name
ORDER BY sport_name;
```

From this we got that below sports have lost their popularity because of significantly less participants over the decades.

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Aeronautics
Alpinism
Art Competitions
Basque Pelota
Cricket
Croquet
Jeu De Paume
Lacrosse
Military Ski Patrol
Motorboating
Polo
Racquets
Roque
Rugby
Tug-Of-War
Motorboating
Polo
Racquets
Roque
Rugby
Tug-Of-War

6. Are there any sports that are specific to a particular region or culture?

To achieve this problem statement, I had to join the sports, events, competitor event, games competitor, person region, and Noc region table through related columns followed by grouping and sorting by region name having distinct sports count to 1.

```
SELECT
    nr.region_name,
    MIN(s.sport_name) AS sport_name,
    COUNT(DISTINCT s.sport_name) AS
        distinct_sport_count
    FROM sport s
    JOIN event e ON s.id = e.sport_id
    JOIN competitor_event ce ON e.id = ce.event_id
    JOIN games_competitor gc ON gc.id = ce.competitor_id
    JOIN person_region pr ON pr.person_id = gc.person_id
    JOIN noc_region nr ON nr.id = pr.region_id
    GROUP BY nr.region_name
    HAVING COUNT(DISTINCT s.sport_name) = 1;
```

So, from the above analysis, we can conclude that there is a specific sport called Athletics which is associated with certain regions such as Newfoundland, North Borneo, Somalia, South Sudan, and Turks and Caicos Islands.

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region_name	sport_name	distinct_sport_count
Newfoundland	Athletics	1
North Borneo	Athletics	1
Somalia	Athletics	1
South Sudan	Athletics	1
Turks and Caicos Islands	Athletics	1

7. Are there any sports that have a higher number of events for one gender compared to others?

To achieve this problem statement, I had to join the person, games competitor, event and sports table through related column followed by grouping the result by sport name by equating the female and male event count through having clause. In select statement I had to use multiple logical function to evaluate male and female event count.

```

SELECT
    s.sport_name,
    SUM(CASE WHEN p.gender = 'M' THEN 1 ELSE 0 END) AS
        male_event_count,
    SUM(CASE WHEN p.gender = 'F' THEN 1 ELSE 0 END) AS
        female_event_count
FROM person p
JOIN games_competitor gc ON p.id = gc.person_id
JOIN competitor_event ce ON gc.id = ce.competitor_id
JOIN event e ON e.id = ce.event_id
JOIN sport s ON s.id = e.sport_id
GROUP BY s.sport_name
HAVING male_event_count <> female_event_count

```

So for Basketball, Judo, Football, Tug-Of-War ,Speed Skating ,Cross Country Skiing ,Athletics ,Ice Hockey, Swimming, Badminton, Sailing, Biathlon, Gymnastics, Art Competitions, Alpine Skiing, Handball, Weightlifting, Wrestling, Luge, Water Polo,Hockey,Rowing,Bobsleigh,Fencing,Equestrianism,Shooting,Boxing,Taekwondo,Cycling,Diving,Canoeing,Tennis,Modern Pentathlon ,Figure Skating ,Golf, Softball, Archery, Volleyball, Synchronized Swimming, Table Tennis, Nordic Combined, Baseball, Rhythmic Gymnastics, Freestyle Skiing, Rugby Sevens, Beach Volleyball, Triathlon, Ski Jumping, Curling, Snowboarding, Rugby, Short Track Speed Skating, Skeleton, Lacrosse, Polo, Cricket, Racquets, Military Ski Patrol, Croquet, Jeu De Paume, Roque, Alpinism, Motorboating, Basque Pelota, Aeronautics no of events are varying.

sport_name	male_event_count	female_event_count
Basketball	3078	1256
Judo	2668	1093
Football	5353	1009
Tug-Of-War	125	0
Speed Skating	3499	2077
Cross Country Skiing	5704	3385

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Athletics	25790	11571
Ice Hockey	4711	754
Swimming	12869	9727
Badminton	717	740
Sailing	5347	925
Biathlon	3024	1863
Gymnastics	16451	9012
Art Competitions	2770	301
Alpine Skiing	5325	3387
Handball	2243	1387
Weightlifting	3262	463
Wrestling	6346	304
Luge	1096	377
Water Polo	3154	488
Hockey	3748	1458
Rowing	7828	2193
Bobsleigh	2823	143
Fencing	8036	1969
Equestrianism	4892	1242
Shooting	8898	1722
Boxing	5648	72
Taekwondo	307	299
Cycling	8776	1394
Diving	1531	1187
Canoeing	4749	1376
Tennis	1585	1146
Modern Pentathlon	1460	164
Figure Skating	1104	1153
Golf	164	67
Softball	0	478
Archery	1244	1005
Volleyball	1847	1543
Synchronized Swimming	0	903
Table Tennis	999	949
Nordic Combined	1329	0
Baseball	891	0
Rhythmic Gymnastics	0	658
Freestyle Skiing	504	433
Rugby Sevens	151	148
Beach Volleyball	287	276

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Triathlon	266	263
Ski Jumping	2357	30
Curling	240	221
Snowboarding	520	416
Rugby	111	0
Short Track Speed Skating	773	759
Skeleton	130	66
Lacrosse	36	0
Polo	89	0
Cricket	18	0
Racquets	12	0
Military Ski Patrol	19	0
Croquet	11	4
Jeu De Paume	11	0
Roque	3	0
Alpinism	15	1
Motorboating	14	1
Basque Pelota	2	0
Aeronautics	1	0

8. Are there any new events that have been introduced in recent editions of the Olympics?

To achieve this problem statement, I had to join the games competitor, competitor event, games and event with related column followed by grouping by event name then I wrapped the joined table in a CTE then extracted the required information.

```

WITH event_years AS (
    SELECT
        e.event_name,
        MIN(g.games_year) AS first_year
    FROM games_competitor gc
    JOIN competitor_event ce ON gc.id = ce.competitor_id
    JOIN games g ON g.id = gc.games_id
    JOIN event e ON e.id = ce.event_id
    GROUP BY e.event_name
)

SELECT event_name, first_year
FROM event_years
WHERE first_year = 2016;

```

So, there are some events which are newly introduced in the recent edition of Olympics games like Rugby Sevens Women's Rugby Sevens, Wrestling Women's Light-Heavyweight, Freestyle, Wrestling Women's Featherweight, Freestyle, Rugby Sevens Men's Rugby Sevens, Sailing Women's Skiffs

event_name	first_year
Rugby Sevens Women's Rugby Sevens	2016
Wrestling Women's Light-Heavyweight, Freestyle	2016
Wrestling Women's Featherweight, Freestyle	2016
Rugby Sevens Men's Rugby Sevens	2016
Sailing Women's Skiff	2016

Sports Analysis (Olympic Games)

9. Are there any events that have been discontinued or removed from the Olympics?

To achieve this problem statement, I had to join the games competitor, competitor event, games and event with their related columns followed by grouping the result by event name then wrapped the joined table in a CTE then extracted the required information.

```
(WITH GAME_YEAR AS
  (SELECT
    e.event_name,
    MIN(g.games_year) AS first_year
  FROM games_competitor gc
  JOIN competitor_event ce ON gc.id = ce.competitor_id
  JOIN games g ON g.id = gc.games_id
  JOIN event e ON e.id = ce.event_id
  GROUP BY e.event_name)
  SELECT event_name ommited_event, first_year
  from GAME_YEAR
  where first_year<>2016
  order by first_year desc;
```

So, yes there are multiple events that have been omitted or removed from the recent edition of Olympics as listed in the table. I have attached some portion of the table to showcase the information.

Ommited_Events_From_Recent_Olympic

Figure Skating Mixed Team
Biathlon Mixed 2 x 6 kilometres and 2 x 7.5 kilometres Relay
Freestyle Skiing Men's Slopestyle
Freestyle Skiing Women's Halfpipe
Ski Jumping Women's Normal Hill, Individual
Snowboarding Women's Slopestyle
Snowboarding Men's Parallel Slalom
Luge Mixed Team Relay
Freestyle Skiing Women's Slopestyle
Snowboarding Men's Slopestyle
Freestyle Skiing Men's Halfpipe
Snowboarding Women's Parallel Slalom
Boxing Women's Flyweight
Sailing Men's One Person Heavyweight Dinghy
Canoeing Men's Kayak Doubles, 200 metres
Canoeing Women's Kayak Singles, 200 metres
Boxing Women's Middleweight
Sailing Men's Skiff
Cycling Men's Omnium
Boxing Women's Lightweight
Cycling Women's Team Pursuit

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10. Are there any notable trends in the height and weight of participants over time

To achieve this problem statement, I had to join the games, games competitor, and person with their related column followed by filtering the result where height and weight are not null then grouping and order the result by games year.

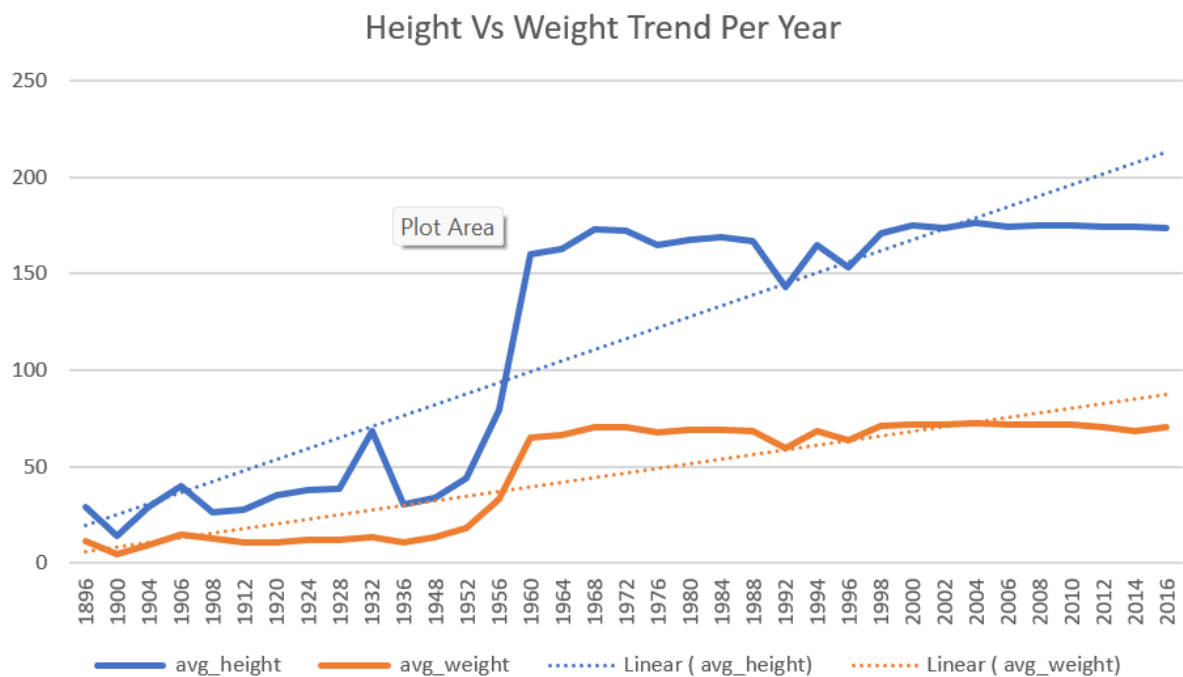
```
select g.games_year, avg(p.height) avg_height, avg(p.weight) avg_weight
      from games g
    join games_competitor gc
      on g.id=gc.games_id
    join person p
      on p.id=gc.person_id
  where p.height is not null and p.weight is not null
    group by g.games_year
    order by g.games_year;
```

Both average height and weight show a clear upward trend over time. The dotted trendlines confirm a positive linear progression. Athletes have generally become taller and heavier over the decades.

games_year	avg_height	avg_weight
1896	29.0556	11.5417
1900	14.0409	4.3947
1904	29.0585	9.3609
1906	39.8875	14.7366
1908	26.4526	12.8149
1912	27.4944	10.5416
1920	35.4965	10.7258
1924	38.2198	11.9502
1928	38.851	12.4053
1932	68.7471	13.4039
1936	30.4394	10.9045
1948	33.5462	13.773
1952	43.98	18.3368
1956	79.493	32.997
1960	160.3199	64.8788
1964	162.8673	66.5791
1968	172.7311	70.6987
1972	172.559	70.3944
1976	164.7108	67.5996
1980	167.3629	68.999
1984	168.8772	68.8856
1988	167.1785	68.6844
1992	143.3956	59.3168
1994	165.2412	68.5251
1996	153.4534	63.4508

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1998	171.1789	70.9931
2000	175.2902	71.7961
2002	173.9165	72.1248
2004	176.4966	72.4817
2006	174.7477	71.9226
2008	174.8395	71.6534
2010	175.2674	72.0032
2012	174.4321	70.3567
2014	174.4225	68.6956
2016	174.0682	70.6179



11. Are there any dominant countries or regions in specific sports or events?

To achieve this problem statement, I had to join the Noc region, person, games competitor, competitor event, medal and sports with their related column followed by grouping the result by region name, event name and sport name then sorting by medal count and limiting the result only for top 10 result.

```
select nr.region_name,e.event_name,s.sport_name, count(ce.medal_id) medal_count
      from noc_region nr
      join person_region pr
      on nr.id=pr.region_id
      join games_competitor gc
      on pr.person_id=gc.person_id
      join competitor_event ce
      on ce.competitor_id=gc.id
      join event e
      on e.id=ce.event_id
      join medal m
      on m.id=ce.medal_id
      join sport s
      on s.id=e.sport_id
  group by nr.region_name,e.event_name,s.sport_name
  order by medal_count desc
  limit 10;
```


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So, yes there are some specific countries which are very dominant in specific event and sports like Sweden bagged maximum medals in Ice Hockey section, India tops in medal count in men's hockey, and USA bagged maximum medals in Art Competitions.

region_name	event_name	sport_name	medal_count
Sweden	Ice Hockey Men's Ice Hockey	Ice Hockey	386
USA	Ice Hockey Men's Ice Hockey	Ice Hockey	376
Canada	Ice Hockey Men's Ice Hockey	Ice Hockey	368
Finland	Ice Hockey Men's Ice Hockey	Ice Hockey	323
India	Hockey Men's Hockey	Hockey	307
UK	Hockey Men's Hockey	Hockey	303
USA	Art Competitions Mixed Painting, Unknown Event	Art Competitions	296
Czechoslovakia	Ice Hockey Men's Ice Hockey	Ice Hockey	291
Switzerland	Ice Hockey Men's Ice Hockey	Ice Hockey	279
Netherlands	Hockey Men's Hockey	Hockey	274

12. What factors contribute to the success or performance of participants from different countries?

To achieve this problem statement, I had to join the Noc region , person region, games competitor, competitor event, event, and person by their related column followed by grouping the result by region name and sorting through medal count in descending order.

```
select nr.region_name,
count(distinct gc.id)total_participants,
count(distinct e.sport_id) sports_count,
count(ce.medal_id) medal_count,
avg(p.height) avg_height,
avg(p.weight) avg_weight
from noc_region nr
join person_region pr
on nr.id=pr.region_id
join games_competitor gc
on gc.person_id=pr.person_id
join competitor_event ce
on ce.competitor_id=gc.id
join event e
on e.id=ce.event_id
join person p
on p.id=pr.person_id
group by nr.region_name
order by medal_count desc;
```

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The analysis indicates that Olympic success is influenced by multiple factors, including athlete participation, sport diversity, and performance efficiency. While countries with larger delegations tend to win more medals overall, smaller nations often demonstrate higher efficiency in terms of medals per athlete. Additionally, participation across a broader range of sports increases medal opportunities. Physical attributes such as height and weight show limited direct correlation with overall success, suggesting that strategic focus, training quality, and specialization play a more significant role in determining performance.

region_▼	total_particip▼	sports_count▼	medal_count▼	avg_height▼	avg_weight▼
USA	12392	58	18127	141.4432	55.9176
UK	7999	59	11536	121.2031	48.5269
France	7323	57	11475	127.1771	48.9933
Germany	7119	53	10496	139.59	56.5142
Italy	6958	51	10355	136.5655	53.0769
Canada	6332	53	9540	149.2739	59.3966
Sweden	5416	47	8335	116.1969	46.9073
Japan	5355	52	8331	152.5606	56.9393
Australia	5452	51	7584	158.092	63.7766
Hungary	4045	43	6514	128.9467	52.5774
Poland	4241	45	6209	162.8156	65.9113
Soviet Union	4263	37	6171	158.001	64.4168
Netherlands	4035	43	5880	117.2073	47.633
Switzerland	3603	51	5832	126.5905	50.5859
Russia	3828	48	5563	168.499	67.5224
Finland	3418	40	5473	142.6983	57.3309
Spain	3867	47	5312	145.868	57.7107
China	3594	47	5133	164.8206	63.451
Austria	3269	46	5016	115.8682	47.6748

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13. Are there any countries that consistently perform well in multiple Olympic editions

To achieve this problem statement, I had to join games, games competitor, person region , Noc region and competitor event through related column followed by grouping the result by region name and games year with required function in select statements.

```
select nr.region_name,
       g.games_year,
       count(distinct gc.id)participant_count,
       count(ce.medal_id) medal_count
  from games g
 join games_competitor gc
    on g.id=gc.games_id
 join person_region pr
    on pr.person_id=gc.person_id
 join noc_region nr
    on nr.id=pr.region_id
 join competitor_event ce
    on ce.competitor_id=gc.id
 group by nr.region_name, g.games_year
 order by medal_count desc
 limit 50;
```

Here top 50 result with respect to medal tally is fetched by considering country and game year and we find that USA tops the list with grabbing maximum medal in multiple Olympics Edition. Also Germany, UK, and RUSSIA also perform well in multiple editions.

region_name	games_year	participant_count	medal_count
USA	1992	685	957
USA	1988	624	886
USA	1904	414	882
Germany	1992	581	866
Unified Team	1992	601	861
USA	1996	646	834
USA	1984	613	824
Soviet Union	1988	581	798
Australia	2000	620	793
Soviet Union	1980	575	789
USA	2000	593	773
USA	1932	500	765
USA	2008	589	764
China	2008	601	733
USA	2004	536	730
USA	2016	554	718
Sweden	1912	443	713
West Germany	1972	502	708
Soviet Union	1976	489	695
Canada	1988	440	694

Sports Analysis (Olympic Games)

West Germany	1984	474	691
UK	2012	532	688
USA	2012	528	686
USA	1976	486	685
USA	1972	476	681
France	1992	451	676
West Germany	1988	437	671
UK	1908	490	670
Canada	1984	475	664
Soviet Union	1972	447	649
Germany	1936	481	640
Canada	1976	442	636
Germany	1996	470	636
Canada	1992	402	629
USA	1968	423	621
Russia	2000	448	620
Germany	1964	432	620
Italy	1992	416	618
Russia	2004	458	612
Russia	2008	463	611
South Korea	1988	423	607
Australia	2004	472	603
UK	1992	420	603
Germany	2004	445	597
Spain	1992	446	595
Soviet Union	1968	386	592
UK	1984	387	587
Brazil	2016	464	585
East Germany	1980	399	582
Germany	2000	427	580

14. Are there any sports or events that have a higher number of medalists from a specific region?

To achieve this problem statement, I had to join Noc region, person region, games competitor, competitor event, event, and sport with their related column followed by grouping the result table by sports name, event name and region name then sorting by medal count in descending order and finally limiting the result to top 10.

Sports Analysis (Olympic Games)

```

select s.sport_name,
       e.event_name,
       nr.region_name,
       count(ce.medal_id) medal_count
from noc_region nr
join person_region pr
on nr.id=pr.region_id
join games_competitor gc
on gc.person_id=pr.person_id
join competitor_event ce
on ce.competitor_id=gc.id
join event e
on e.id=ce.event_id
join sport s
on e.sport_id=s.id
group by s.sport_name,e.event_name,nr.region_name
order by medal_count desc;
limit 10;

```

So, from this analysis we got that particularly in men's ice hockey event, Sweden, Usa, Canada, Finland, Czechoslovakia and Switzerland bagged maximum no of medals.

sport_name	event_name	region_name	medal_count
Ice Hockey	Ice Hockey Men's Ice Hockey	Sweden	386
Ice Hockey	Ice Hockey Men's Ice Hockey	USA	376
Ice Hockey	Ice Hockey Men's Ice Hockey	Canada	368
Ice Hockey	Ice Hockey Men's Ice Hockey	Finland	323
Hockey	Hockey Men's Hockey	India	307
Hockey	Hockey Men's Hockey	UK	303
Art Competitions	Art Competitions Mixed Painting, Unknown Event	USA	296
Ice Hockey	Ice Hockey Men's Ice Hockey	Czechoslovakia	291
Ice Hockey	Ice Hockey Men's Ice Hockey	Switzerland	279
Hockey	Hockey Men's Hockey	Netherlands	274

Sports Analysis (Olympic Games)

15. What are some notable instances of unexpected or surprising medal wins?

To achieve this problem statement, I had to join games, games competitor, person region, Noc region and competitor event through related column followed by grouping by region name and order by game year descending order.

```
select nr.region_name,  
       min(g.games_year) game_yr,  
       count(ce.medal_id) medal_count  
  from games g  
 join games_competitor gc  
    on g.id=gc.games_id  
 join person_region pr  
    on pr.person_id=gc.person_id  
 join noc_region nr  
    on nr.id=pr.region_id  
 join competitor_event ce  
    on ce.competitor_id=gc.id  
 group by nr.region_name  
 order by game_yr desc;
```

So, yes there is a surprise that Refugee Olympic team region participated 1st time in the recent Olympic event and grabbed a notable amount of medal (12 medals).

region_name	game_yr	medal_count
Refugee Olympic Team	2016	12

16. Are there any regions that have experienced significant growth or decline in Olympic participation?

To achieve this problem statement, I had to join the games, games competitor, person region , Noc region and competitor event followed by grouping by region name and games year then wrapping this joined table inside a CTE. Through a window function LAG () I achieved the previous year athlete participants partition by country and order by Olympic year.

country	oly_year	current_yea	prev_year_athlete	athlete_diff
Australia	2000	620	23	597
Australia	2002	25	620	595
Unified Team	1994	62	601	539
USA	1994	148	685	537
China	2008	601	73	528

Sports Analysis (Olympic Games)

```
with participation as
(select nr.region_name country,
g.games_year oly_year,
count(distinct gc.person_id) total_athlete
from games g
join games_competitor gc
on g.id=gc.games_id
join person_region pr
on pr.person_id=gc.person_id
join noc_region nr
on nr.id=pr.region_id
join competitor_event ce
on ce.competitor_id=gc.id
group by nr.region_name,g.games_year
order by nr.region_name,g.games_year),
trend as
(select country,
oly_year,
total_athlete current_year_athlete,
lag(total_athlete,1) over(partition by country order by oly_year) prev_year_athlete,
abs(total_athlete-lag(total_athlete,1) over(partition by country order by oly_year)) athlete_diff
from participation)

select *
from trend
order by athlete_diff desc
limit 5;
```

So, yes there are multiple countries like Australia, Unified team, USA and China are there which have experienced a notable change in athlete participation count in consecutive year.

17. How do cultural or geographical factors influence the performance of regions in specific sports

To achieve this problem statement, I had to join the competitor event, games competitor, person region, Noc region, event, sports and person through related column. After that I had to group the table by region name and sport name. Later overall height, overall weight and total participants and BMI was calculated in the select statement to show case the effect.

region_name	sport_name	total_participant	overall_height	overall_weight	BMI	medal_count
USA	Athletics	1837	167.1472	67.4749	24.151496	3104
UK	Athletics	1211	133.643	52.1717	29.2107398	2196
USA	Gymnastics	275	108.9082	35.1235	29.6126058	1830
USA	Swimming	678	152.8828	60.2046	25.7580166	1579
France	Athletics	798	141.002	53.6844	27.002101	1464
France	Gymnastics	201	111.585	39.4852	31.7119324	1383
Germany	Athletics	764	167.661	68.2981	24.2965448	1345
Italy	Gymnastics	204	105.6	34.0007	30.4901824	1335
UK	Swimming	474	142.8044	53.8265	26.3944711	1268
Australia	Swimming	386	158.3227	61.4624	24.520151	1224

Sports Analysis (Olympic Games)

```
select nr.region_name,  
       s.sport_name,  
       count(distinct gc.person_id) total_participant,  
       avg(p.height) overall_height,  
       avg(p.weight) overall_weight,  
       10000*(avg(p.weight)/power(avg(p.height),2)) BMI,  
       count(ce.medal_id) medal_count  
from competitor_event ce  
join games_competitor gc  
on ce.competitor_id=gc.id  
join person_region pr  
on pr.person_id=gc.person_id  
join noc_region nr  
on nr.id=pr.region_id  
join event e  
on e.id=ce.event_id  
join sport s  
on s.id=e.sport_id  
join person p  
on p.id=gc.person_id  
group by nr.region_name,s.sport_name  
order by medal_count desc  
limit 10;
```

So, from the result we can say that there are some countries whose participant counts and physical balance in combo are good, grabbed maximum number of medals like USA, UK, FRANCE, GERMANY, ITALY and AUSTRALIA.

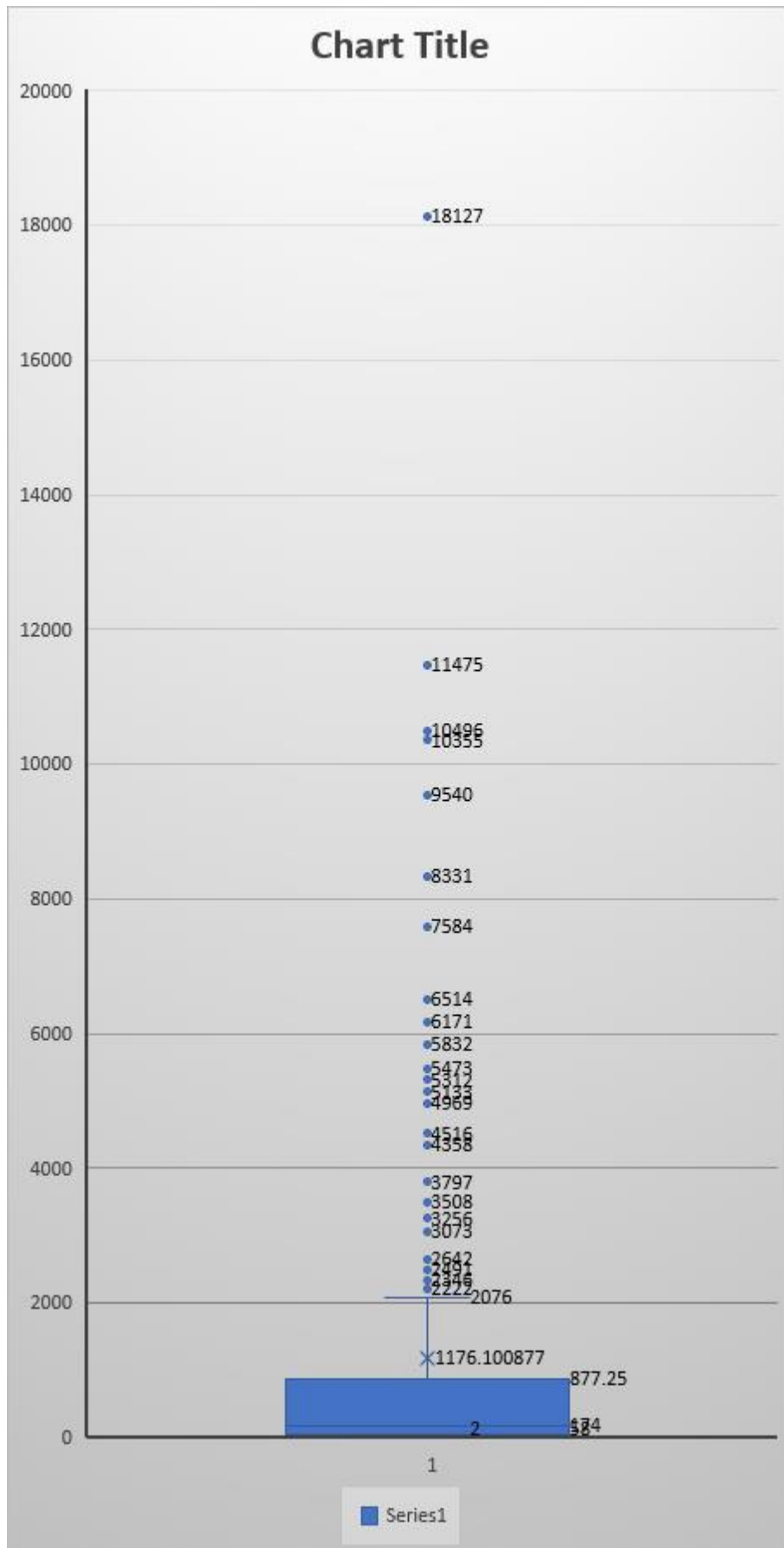
18. Are there any regions that have had a notable impact on the overall medal tally?

To achieve this problem statement, I had to join the competitor event, games competitor, person region, noc region and event followed by grouping the table by region name and sorting by total medal in descending order.

```
select nr.region_name country,  
       count(ce.medal_id) total_medal  
from competitor_event ce  
join games_competitor gc  
on ce.competitor_id=gc.id  
join person_region pr  
on pr.person_id=gc.person_id  
join noc_region nr  
on nr.id=pr.region_id  
join event e  
on e.id=ce.event_id  
group by nr.region_name  
order by total_medal desc;
```


Sports Analysis (Olympic Games)

Then I have created a box plot to find out all the outliers who impacted the overall medal tally.



Sports Analysis (Olympic Games)

So, there are some countries who impacted heavily in the overall medal tally which is around 1176 because there are 36 data points (medal count by country) which are being considered as outliers from whisker plot. These countries are listed below.

Impactful_region

USA

UK

France

Germany

Italy

Canada

Sweden

Japan

Australia

Hungary

Poland

Soviet Union

Netherlands

Switzerland

Russia

Finland

Spain

China

Austria

Norway

Czechoslovakia

South Korea

Romania

West Germany

Brazil

Denmark

Bulgaria

Belgium

East Germany

Argentina

Ukraine

Mexico

Yugoslavia

Cuba

New Zealand

Greece

Sports Analysis (Olympic Games)

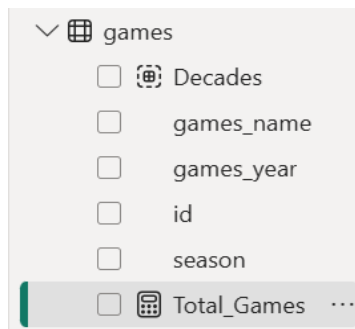
PowerBI Dashboard:

Problem Statements:

1.How many Olympic Games have been held in each season (Summer vs. Winter)?

To achieve this problem statement, I had to create a measure called “Total_games” then placed this measure inside the games table in the data pane. Then I created a KPI card and called this measure inside it. After that I have formatted the look and feel of the KPI card for visually appealing.

```
Total_Games = DISTINCTCOUNT(games[id])
```



A slicer is selected from visualization pane added to report page for showcasing the game count as per each season.



The KPI card is adjusted according to season slicer and we got that there are 29 Olympics games were held during Summer and 22 games were held during Winter.

2.What is the distribution of games across different decades?

To achieve this problem statement first of all I grouped the games year in games table according to decade then from the visualisation pane I selected a simple table and called both the games year(grouped) and total games measure inside it.

Decades	Total_Games
1918-1928	2
1929-1939	2
1940-1950	1
1951-1961	3
1962-1972	3
1973-1983	2
1984-1994	4
1995-2005	2
2006-2016	3
Total	22

A screenshot of the Power BI data pane. The 'games' table is expanded, showing fields: Decades, games_name, games_year, id, season, and Total_Games. The 'Decades' field is highlighted with a blue bar on the left.

Sports Analysis (Olympic Games)

3. Which cities have hosted the most Olympic Games?

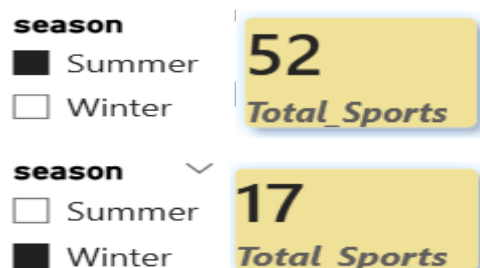
To achieve this problem statement, I selected a simple table from the visualisation pane then called the city name column from city table in data pane and total games measure inside it and then sorting the total games column in descending manner. Then the games count per city is fetched.

city_name	Total_Games
Athina	3
London	3
Innsbruck	2
Lake Placid	2
Los Angeles	2
Paris	2
Sankt Moritz	2
Stockholm	2
Albertville	1
Amsterdam	1
Antwerpen	1
Atlanta	1
Total	51

4. What is the distribution of sports between Summer and Winter Olympics?

To achieve this problem statement, I had to create a Total Sports measure and placed it inside sports table in data pane. A KPI card is selected from visualisation pane and the measure is called inside it. After that a slicer is selected from the visualisation pane and called the season column inside it and with respect to slicer the value of the KPI card adjusted.

`Total_Sports = DISTINCTCOUNT(sport[id])`



So we got that there is 52 sports were scheduled during Summer Olympics and 17 Sports were scheduled during winter Olympics.

Sports Analysis (Olympic Games)

5. Which sports have the highest number of events in the Olympics?

To achieve this problem statement, I created a total event measure and placed it under the event table in data pane. Then a simple table is selected from the visualisation section then called both the total event measure and sports name from sports column inside this table visual. Then I achieved the required information by changing the sorting arrow inside table visual and found that Athletics have the highest number of events in the Olympics.

```
Total_Events = DISTINCTCOUNT(event[id])
```

sport_name	Total_Events
Athletics	83
Shooting	83
Swimming	54
Cycling	44
Sailing	37
Wrestling	30
Art Competitions	29
Canoeing	27
Gymnastics	27
Archery	25
Cross Country Skiing	23
Rowing	23
Total	757

757
Total_Events

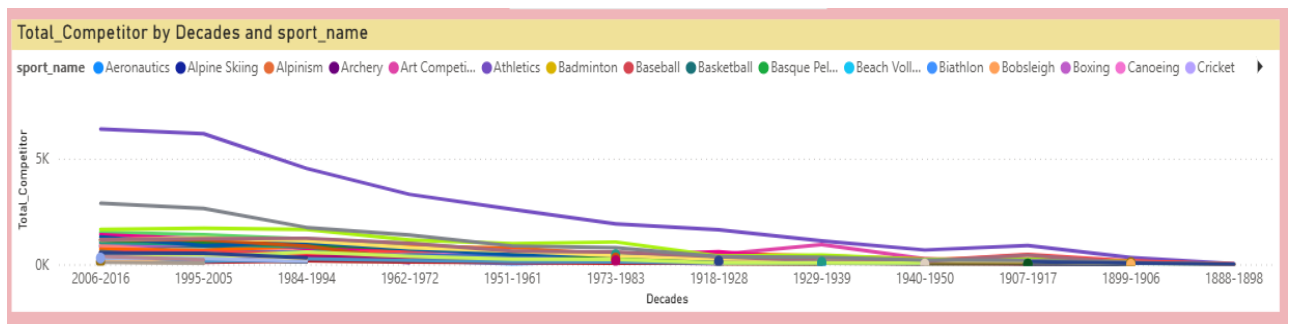
6. How has the participation in each sport evolved over time?

To achieve this problem statement, I created a total competitor measure first and placed it inside the game competitor table in data pane. Then I selected a line chart visual to showcase the decade wise trend. After that I called the total competitor measure inside Y axis of the visual, decade in the X-axis and put sports name as legend. Then from the visual we can clearly get the information regarding sports participants evolved over time.

```
Total_Competitor = DISTINCTCOUNT(competitor_event[competitor_id])
```

X-axis
Decades
Y-axis
Total_Competitor

Sports Analysis (Olympic Games)



7.How many events are there in each sport?

To achieve this problem statement, I created a total event measure and placed it under the event table in data pane. Then a simple table is selected from the visualisation section then called both the total event measure and sports name from sports column inside this table visual. Then I achieved the required information.

`Total_Events = DISTINCTCOUNT(event[id])`

sport_name	Total_Events
Athletics	83
Shooting	83
Swimming	54
Cycling	44
Sailing	37
Wrestling	30
Art Competitions	29
Canoeing	27
Gymnastics	27
Archery	25
Cross Country Skiing	23
Rowing	23
Total	757

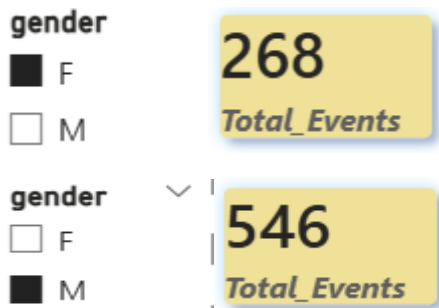
757

Total_Events

Sports Analysis (Olympic Games)

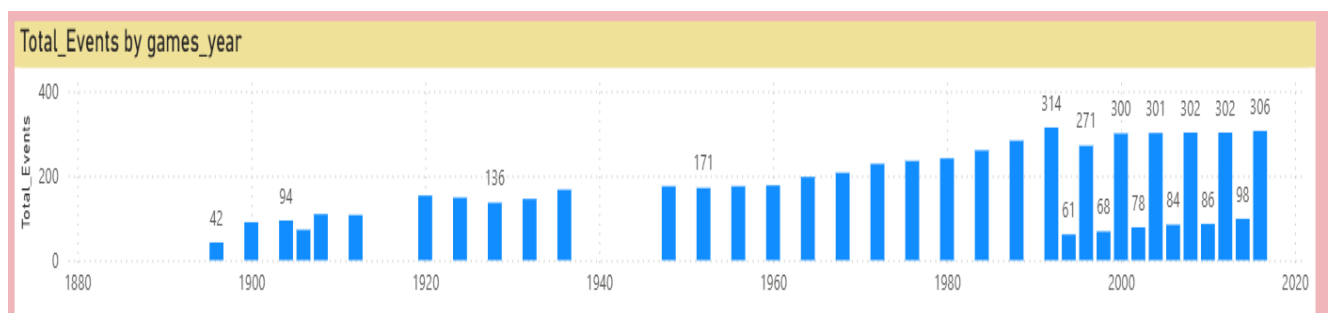
8.What is the distribution of events by gender (Men, Women, Mixed)?

To achieve this problem statement, I created total event measure, placed under event table in data pane, selected a KPI card and called the measure inside it. Then a slicer is selected from the visualization pane and according to the slicer the total event KPI card showed case the result as 268 events were for female and 546 events were for male.



9.How has the number of events changed over time?

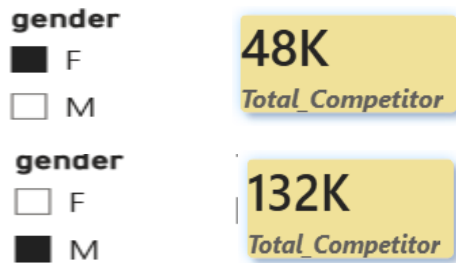
To achieve this problem statement, I selected a column chat from visualisation pane and called both the Total event measure and games year column from games table inside this column visual to show case the result.



Sports Analysis (Olympic Games)

10.What is the distribution of participants by gender?

To achieve this problem statement, I selected a KPI card visual from the visualisation pane and called the total competitor measure inside it. Then a slicer is selected from the visuals and gender column from the person table is called inside it. So according to slicer selection the required information is obtained as 48000 participants are under female category and 132000 participants are under male category.



11.Which countries have the highest number of participants in the Olympics?

To achieve this problem statement, I selected a table visual from the visual pane then called the total competitor measure and region name(country) column from noc region table from the data pane inside the selected table visual. Then after changing the sorting arrow for the value column inside the table visual we found that USA has the highest number of participants in Olympics history as 12392.

Columns	
sport_name	▼ ✕
Total_Events	▼ ✕

region_name	Total_Competitor
USA	12392
UK	7999
France	7323
Germany	7119
Italy	6958
Canada	6332
Australia	5452
Sweden	5416
Japan	5355
Soviet Union	4263
Poland	4241
Hungary	4045

Sports Analysis (Olympic Games)

12.How does the age distribution of participants vary across different games?

To achieve this problem statement, I had to create a age group by using the age column in games competitor table with 10 years gap like 10-19, 20-29, 30-39 etc. Then I selected a clustered column visual from the visual pane. The total participant measure was called inside Y axis, game names column is called inside X axis with age group column as legend and we can clearly see the distribution result.

X-axis

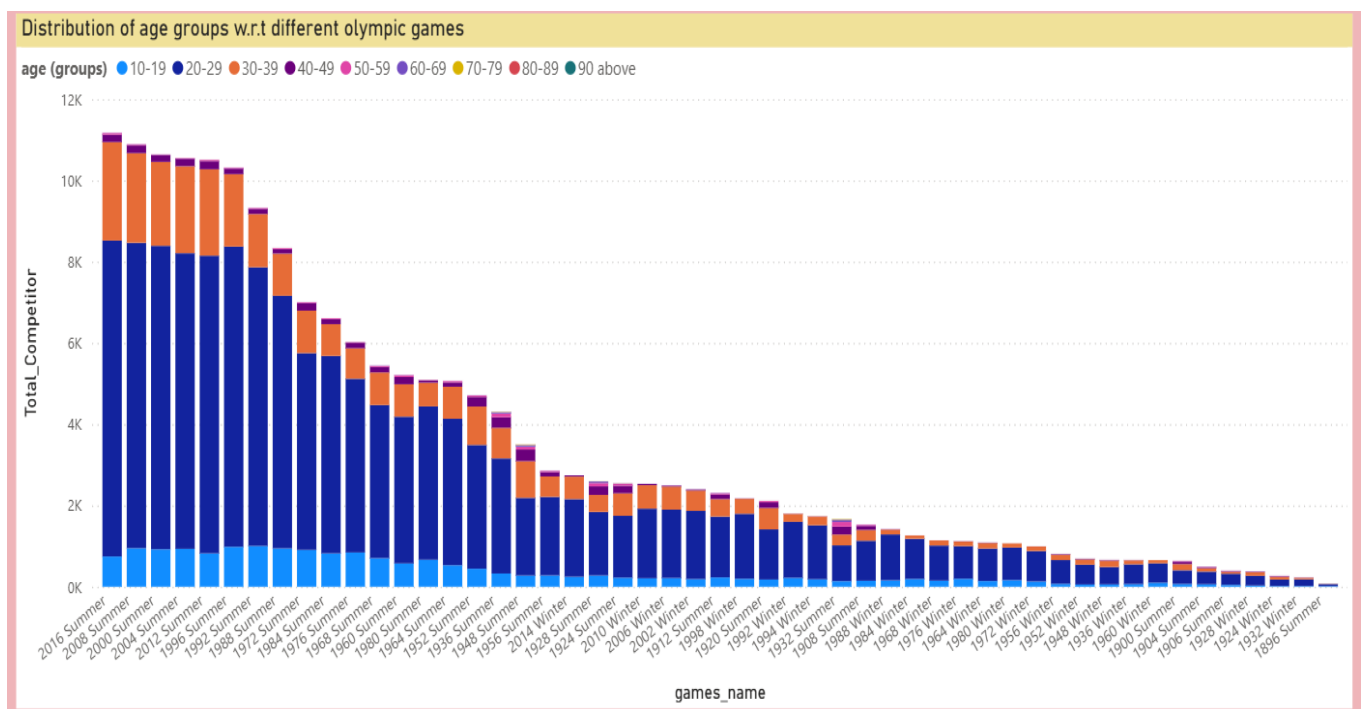
games_name

Y-axis

Total_Competitor

Legend

age (groups)



Sports Analysis (Olympic Games)

13.How many medals have been awarded in each Olympics?

To achieve this problem statement, I created a total medal measure and placed it inside the medal column in data pane. Then I selected a column visual from the visual pane and placed the total medal measure in the Y-axis and games name column inside X-axis then enabling the data level from the visual setting we can get the required information.

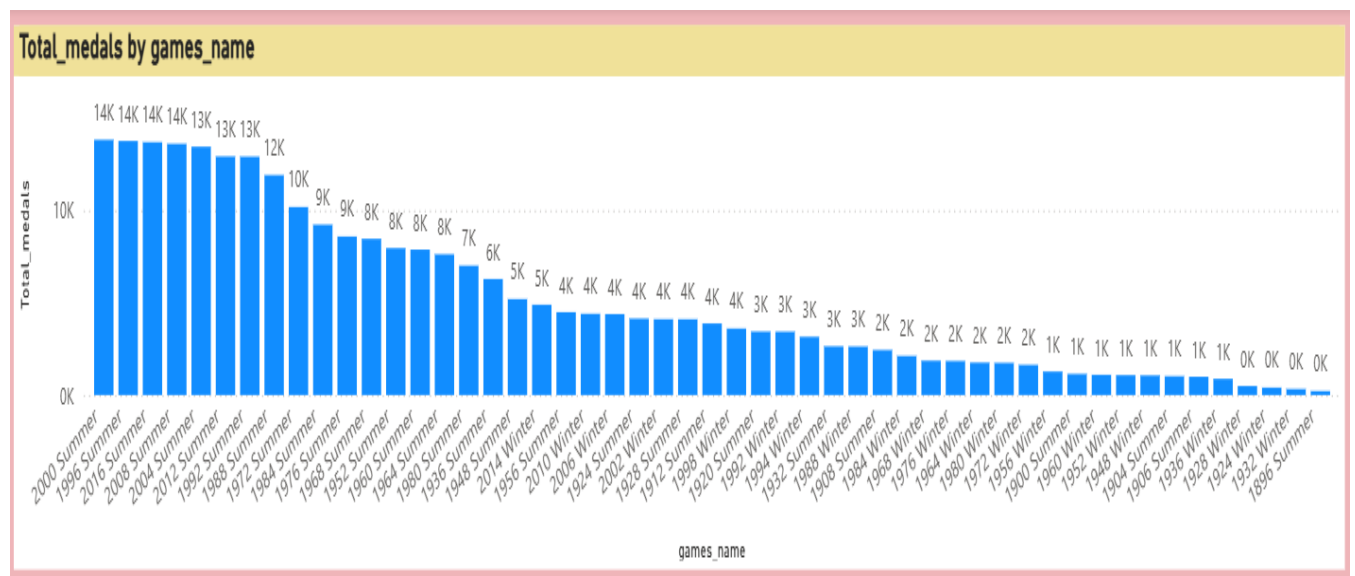
```
Total_medals = count(competitor_event[medal_id])
```

X-axis

games_name

Y-axis

Total_medals



14.Which countries have the highest number of gold medals?

To achieve this problem statement, I created a measure by calculate functionality as Total Gold Awarded then placed it under competitor event table in the data pane then selected a table visual from the visual pane and called region name column from noc region table and Total Gold Awarded measure inside it then sorting the measure column by changing the arrow to descending order.

Sports Analysis (Olympic Games)

Then in result we got that USA received highest gold medals as 2533.

```
Total_Gold_Awarded = CALCULATE(COUNT(competitor_event[medal_id]),competitor_event[medal_id]=1)
```

region_name

Total_Gold_Awarded

region_name	Total_Gold_Awarded
USA	2533
Soviet Union	1135
Germany	843
UK	658
Italy	568
France	492
Sweden	477
Russia	466
Canada	461
East Germany	451
Hungary	434
Norway	378
China	350
Australia	348

15.How does the medal distribution vary across different sports?

To achieve this problem statement, I selected a pie chat from the visual pane and called the Total Medal measure and sports name column into the value and legend section of the visual respectively. Then we clearly got the distribution.

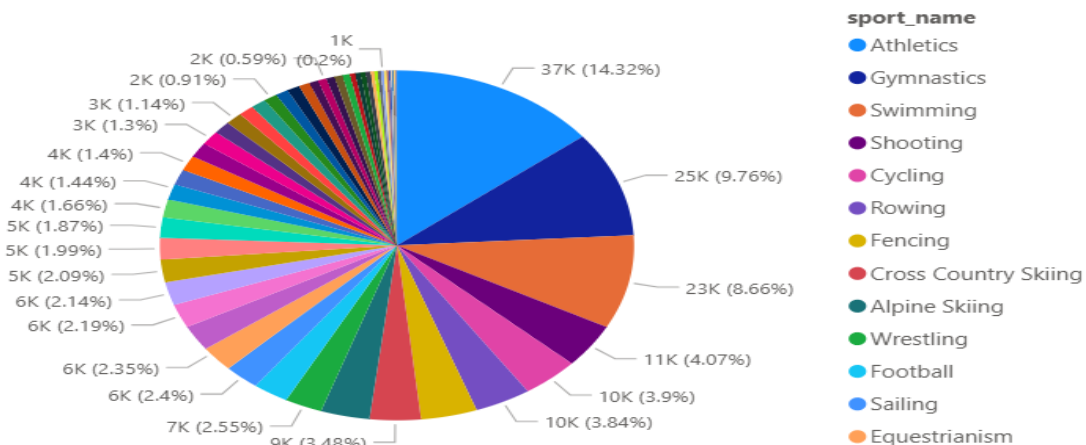
Legend

sport_name

Values

Total_medals

Total_medals by sport_name



Sports Analysis (Olympic Games)

16.How many regions or NOCs participate in each Olympic Games?

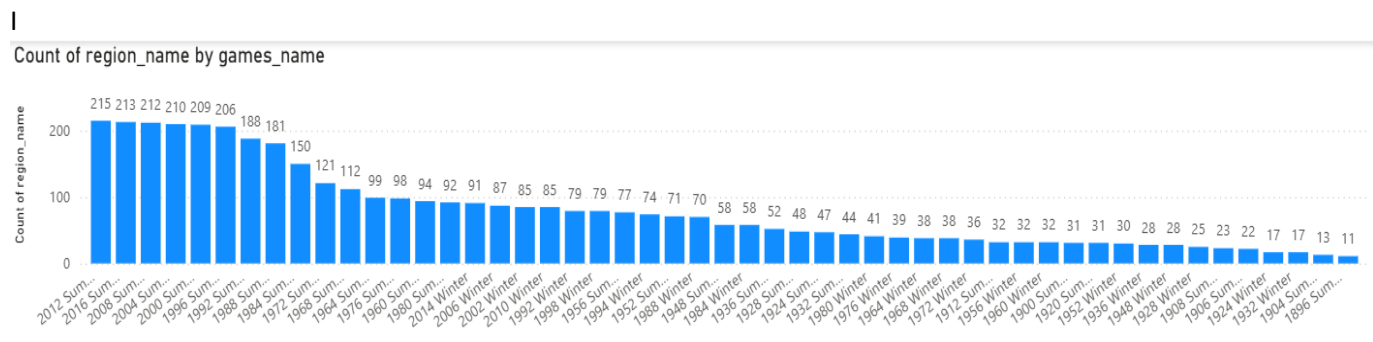
To achieve this problem statement, I selected a column chart from the visual pane and called the games name column and region name column inside X-axis and Y-axis respectively. So we can clearly show case the no of countries or Noc Regions participated in each Olympics game season.

X-axis

games_name

Y-axis

Count of region_name



17.Which regions have the highest number of participants in the Olympics?

To achieve this problem statement, I selected a table visual from the visual pane and called the Total Participants measure and region name column inside it. Then changing the sorting arrow for the measure column for descending order we got that USA has the highest number of participants in the Olympics.

region_name	Total_Competitor
USA	12392
UK	7999
France	7323
Germany	7119
Italy	6958
Canada	6332
Australia	5452
Sweden	5416
Japan	5355
Soviet Union	4263
Poland	4241
Hungary	4045

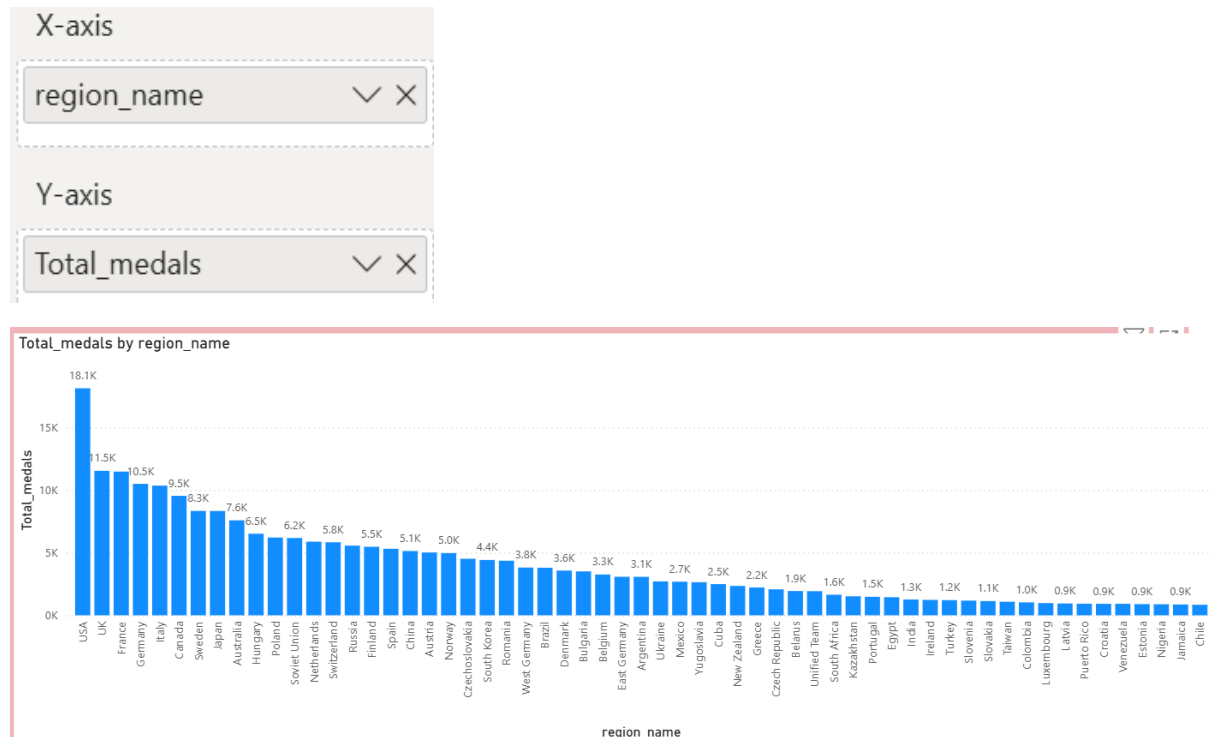
region_name

Total_Competitor

Sports Analysis (Olympic Games)

18.What is the distribution of medals among different regions?

To achieve this problem statement, I selected a column chart from the visual pane and called the Total medal measure and region name column inside this visual to show case the result.



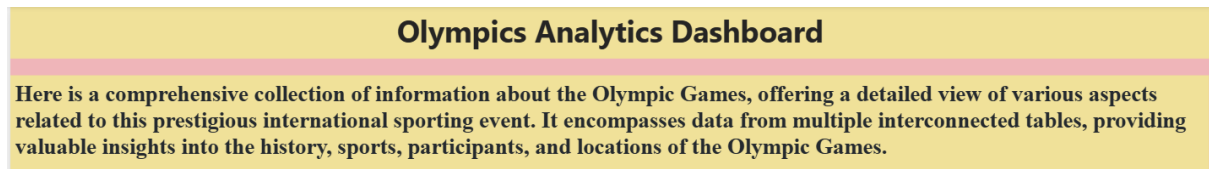
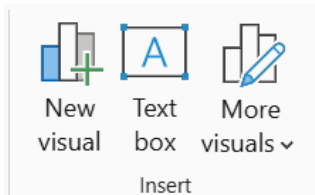
Final Interactive Dashboard:

First of all, I went to model view and in manage relationship section made all the relationships active and enabled the both direction filter from **Dimension Table --- Data Table** and **Data Table -- Dimention Table**.

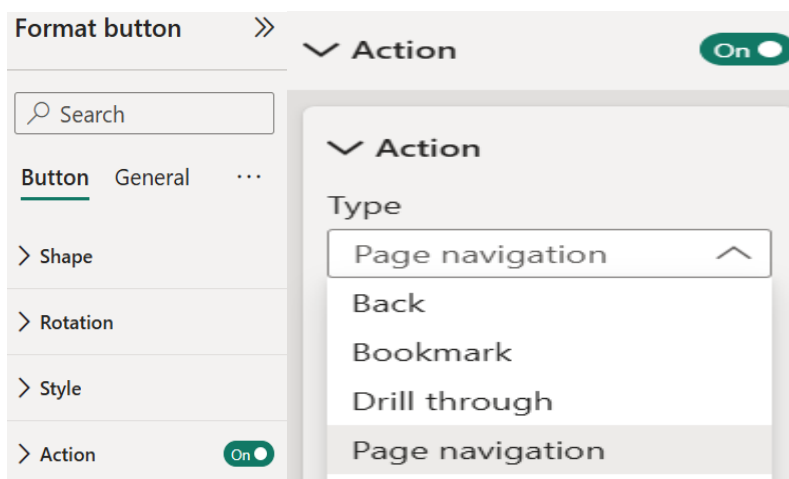
From: table (column) ↑	Relationship	To: table (column)	Status
competitor_event (competitor...		games_competitor (id)	Active
competitor_event (event_id)		event (id)	Active
competitor_event (medal_id)		medal (id)	Active
event (sport_id)		sport (id)	Active
games_city (city_id)		city (id)	Active
games_city (games_id)		games (id)	Active
games_competitor (games_id)		games (id)	Active
games_competitor (person_id)		person (id)	Active
person_region (person_id)		person (id)	Active
person_region (region_id)		noc_region (id)	Active

Sports Analysis (Olympic Games)

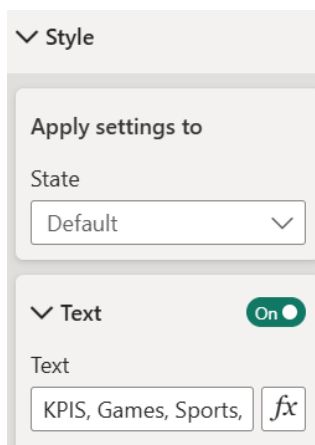
A title box was inserted as text box where I wrote the title as “Olympics Analytics Dashboard” and made some change for the visual appealing context and project overview.



After that I inserted 4 buttons for navigation purpose to directly navigate to required information as per the problem statements. In each button I enabled the action button. And in action I choose page navigator and put the required page name where the relevant information and visuals are there.



Then in the style section of the formatting pane of the selected button, I wrapped some informative text as per problem statements by enabling the text button.



Then 4 action buttons I inserted in the front visual.

Sports Analysis (Olympic Games)

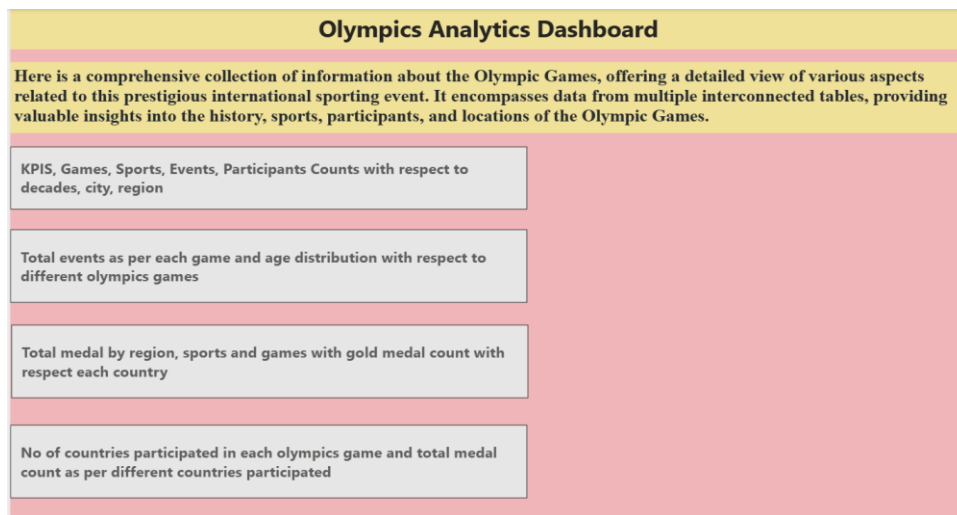
KPIS, Games, Sports, Events, Participants Counts with respect to decades, city, region

Total events as per each game and age distribution with respect to different olympics games

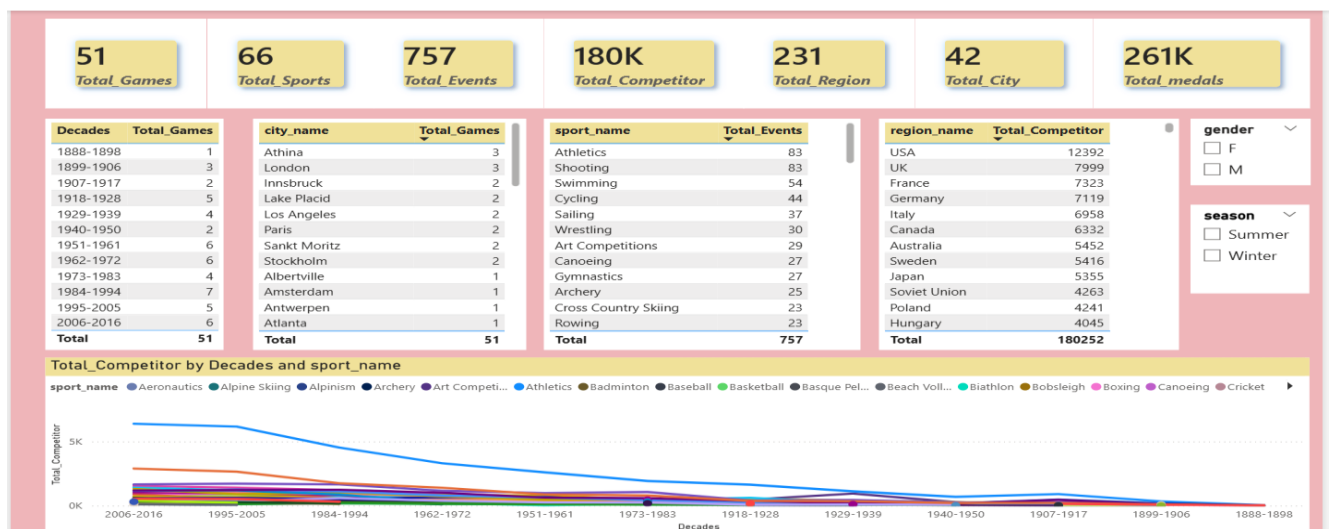
Total medal by region, sports and games with gold medal count with respect each country

No of countries participated in each olympics game and total medal count as per different countries participated

Then all the KPIS, charts, graphs, trends and visuals are aligned properly to finalise the dashboard which operates interactively.

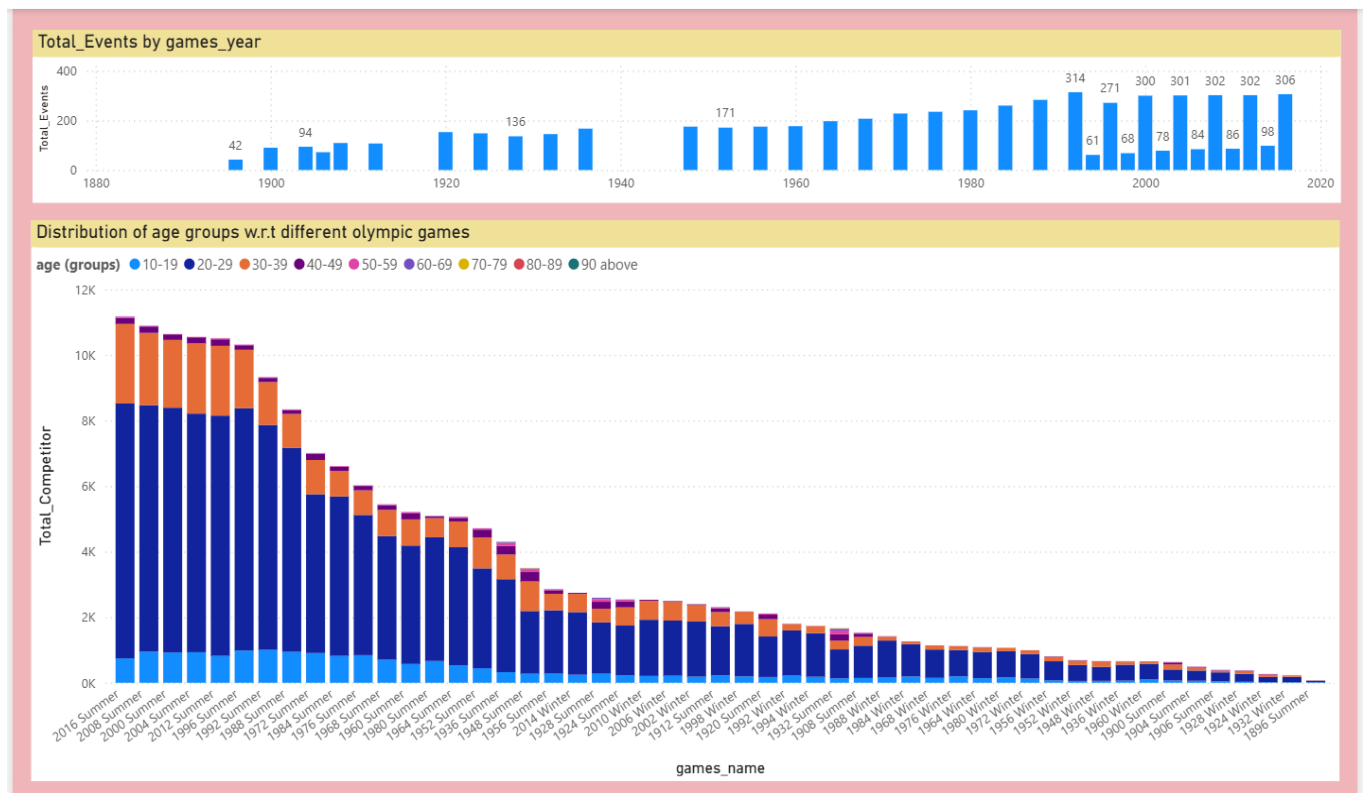


If I click CTRL+ select the 1st action button:

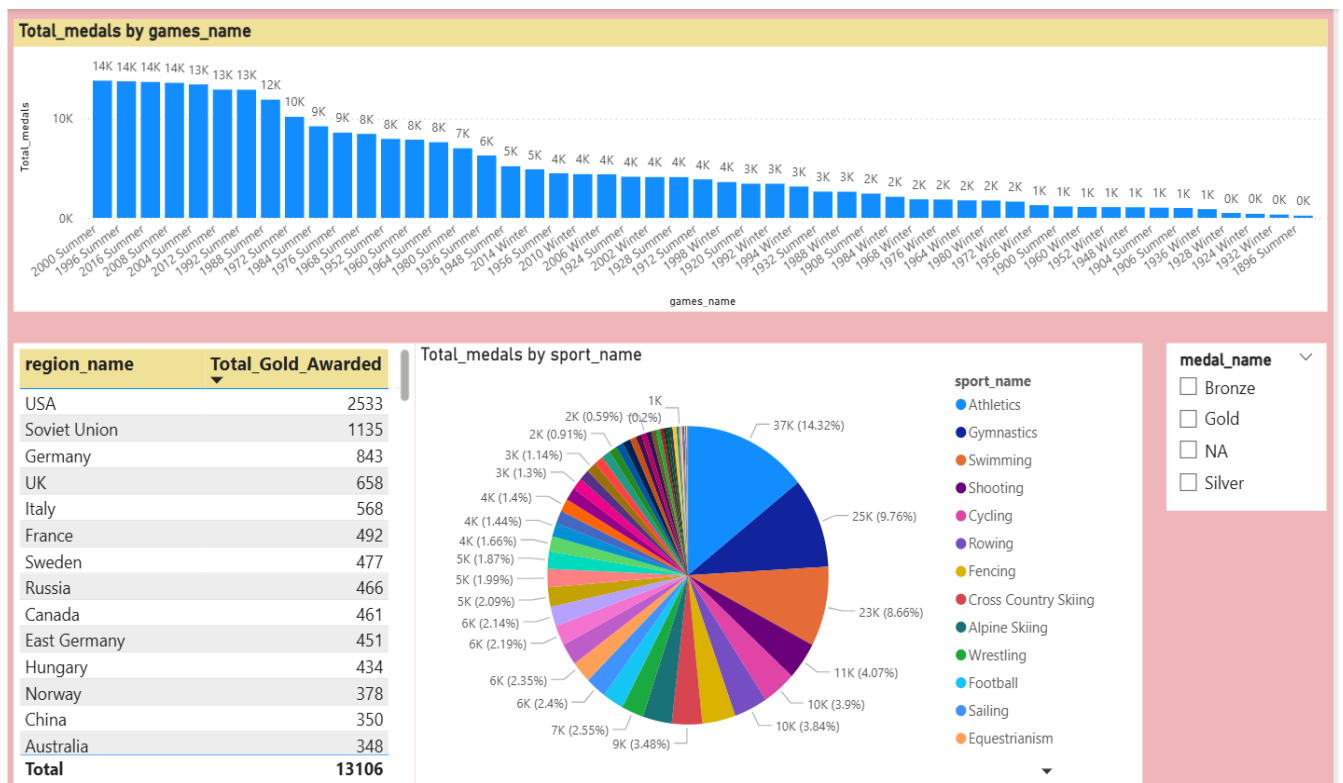


Sports Analysis (Olympic Games)

If I click CTRL+ select the 2nd action button:

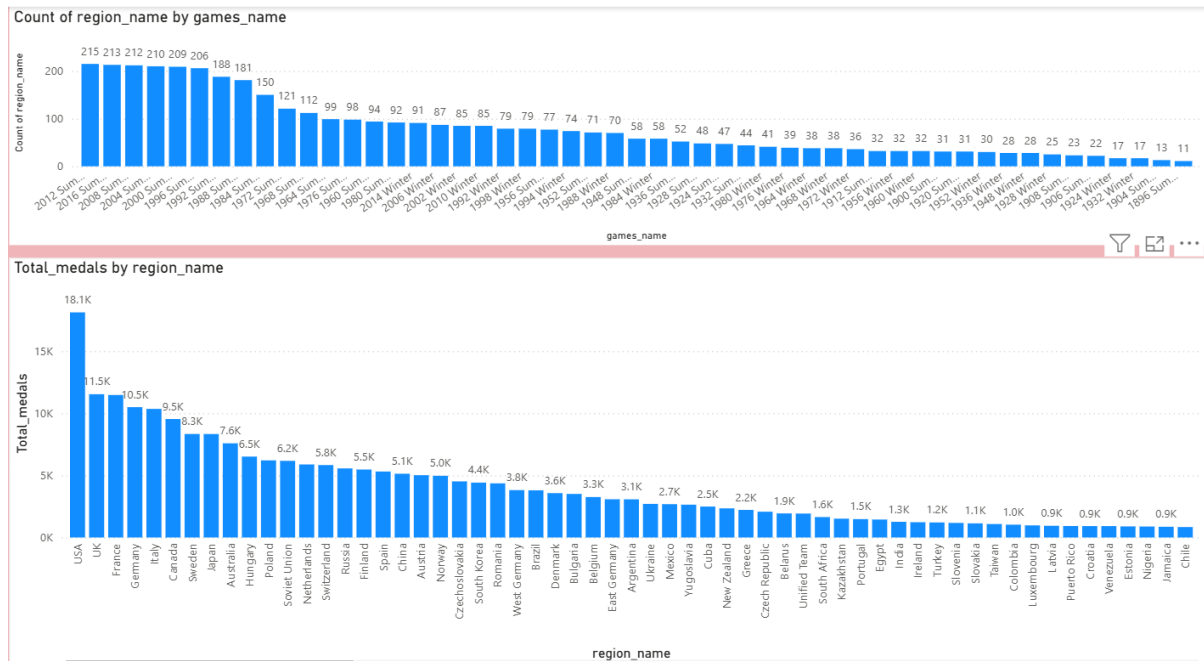


If I click CTRL+ select the 3rd action button:



Sports Analysis (Olympic Games)

If I click CTRL+ select the 3^d action button:



So, our final interactive Olympic Games Dashboard is ready with all the answers for the problem statements.